Flood Extent Mapping Using a Unet-Resnet50 Hybrid Deep Learning Model

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***Abstract*— Accurate flood-extent mapping is essential for rapid disaster response and damage assessment. Deep learning enables automated flood detection, but traditional architectures such as UNet often struggle to learn deep contextual features. This work presents a hybrid UNet–ResNet50 architecture applied to the "Synthetic Flood Imagery for Image Segmentation" dataset. The model integrates a ResNet50 encoder with a UNet decoder for pixel-wise flood segmentation. Experimental results show that the proposed model achieves** a **test IoU of 0.8643**, **F1-score of 0.86, and overall pixel accuracy of 0.86**, **outperforming baseline UNet implementations. The model also achieves a validation IoU of 0.6189 with a validation loss of 0.3793, while the final test loss is 0.3581. These results confirm that hybrid encoder–decoder architectures improve segmentation performance for synthetic flood imagery and are viable candidates for real-time flood mapping systems.**

Keywords— Flood mapping, UNet, ResNet50, semantic segmentation, deep learning, remote sensing.

I. INTRODUCTION

Floods are among the most frequent and destructive natural disasters worldwide, causing severe impacts on human life, infrastructure, and the environment. Rapid identification of inundated regions is essential for emergency response, resource allocation, and long-term planning. Traditional flood-mapping methods rely on spectral index thresholding such as the Normalized Difference Water Index (NDWI) and Modified NDWI (MNDWI) or manual interpreter-based assessment using optical imagery. However,these are highly sensitive to illumination variation, cloud cover, vegetation density, and mixed land–water boundaries, resulting in unreliable outputs in complex environments [1], [2]. The emergence of deep learning, particularly convolutional neural networks (CNNs), has significantly improved remote-sensing image segmentation. Architectures such as Fully Convolutional Networks (FCN) [3], UNet [4], and DeepLab models [5] have achieved state-of-the-art performance in various segmentation tasks, including biomedical imaging and land-cover classification. Among, UNet is widely used due to its encoder–decoder structure and skip connections, making it highly effective for dense pixel-level prediction. The shallow encoder of classical UNet limits its ability to extract deep semantic features from complex scenes. Residual Networks (ResNet), introduced by He et al. [6], resolve the vanishing gradient problem and allow deeper feature extraction through residual skip connections. Integrating ResNet as the encoder in a UNet framework has shown improved performance in high-level feature extraction while maintaining UNet's structural advantage in localization. Such hybrid models have been successfully applied in medical imaging, urban mapping, and environmental monitoring [7], [8].

Despite these advances, effective flood mapping solutions remain limited by the lack of diverse annotated training datasets. Real flood imagery often requires manual labeling, which is time consuming and sometimes not convienient. Synthetic training datasets, such as Synthetic Flood Imagery for Image Segmentation dataset, provide an alternative by enabling large-scale supervised training under controlled variations.

Recent advances in synthetic data generation have further expanded opportunities for training robust deep learning models with real world annotated datasets are scarce. Synthetic remote sensing datasets allow controlled simulation of flood scenarios with diverse textures, lighting conditions, and water distributions, enabling models to learn generalizable spatial patterns that replicate real inundation behavior [14], [15]. This is particularly beneficial for flood mapping applications where real imagery may be limited due to cloud obstruction, satellite revisit times, or unavailable ground-truth labels. Studies have shown that models trained on synthetic data can achieve strong performance when fine-tuned or transferred to real-world tasks, especially when combined with high capacity architectures such as residual encoders [16]. Leveraging these advantages, the present work employs a synthetic flood dataset to train a powerful UNet–ResNet50 hybrid model capable of robust segmentation and improved feature extraction, demonstrating that synthetic data paired with advanced deep-learning architectures can form an effective foundation for automated flood-mapping systems.

This study proposes a UNet–ResNet50 hybrid deep learning model for flood extent segmentation using synthetic imagery. The model leverages the deep representational strength of ResNet50 and the spatial reconstruction capability of UNet. The hybrid architecture is trained and validated using the provided dataset and achieves strong segmentation performance with a test IoU of 0.8643**,** F1-score of 0.86**,** and pixel accuracy of 0.86. These results demonstrate the potential of encoder & decoder hybrid models for automated flood-mapping applications.

II. LITERATURE REVIEW

Flood mapping has traditionally been performed using spectral indices and threshold-based algorithms applied to optical satellite imagery. Early methods used the Normalized Difference Water Index (NDWI) and its variants to distinguish water bodies from land surfaces [1], [2]. While effective under clear-sky conditions, these index-based approaches are highly sensitive to atmospheric changability, cloud cover, and mixed pixels, limiting their reliability during real flood events. Classical machine learning classifiers such as Random Forest and Support Vector Machines have also been used for water classification [7], but they require handcrafted features and lack adaptability across diverse terrain and flood scenarios.

The introduction of deep learning significantly improved remote sensing–based segmentation tasks. Long et al. introduced Fully Convolutional Networks (FCNs) as the first end-to-end architecture for pixel-wise prediction, establishing the foundation for semantic segmentation [3]. UNet, proposed by Ronneberger et al. [4], enhanced this approach through skip connections between encoder and decoder, allowing better localization and boundary recovery. Its strong performance in medical and remote-sensing applications inspired numerous adaptations and variants. DeepLab architectures further advanced segmentation by incorporating atrous convolutions and multi-scale processing [5], demonstrating improved performance in complex environments.

Residual learning, introduced through ResNet architectures, addressed the degradation problem in deep networks by using identity skip connections, enabling effective training of very deep feature extractors [6]. Several studies have integrated ResNet backbones into encoder–decoder frameworks, significantly improving segmentation quality for land-cover mapping, building extraction, and environmental monitoring tasks [7], [8]. Hybrid models combining residual encoders and UNet decoders have demonstrated superior performance due to their ability to extract deep semantic features while preserving spatial detail.

In the domain of flood mapping, deep learning has become a promising alternative to classical remote-sensing techniques. Jain et al. demonstrated the use of CNNs for flood detection from multispectral imagery [9], while other studies utilized SAR data to overcome cloud limitations [14]. However, research remains constrained by the scarcity of annotated flood datasets, limiting model generalization across real-world flood conditions. Synthetic datasets have recently emerged as a solution to this challenge, enabling scalable supervised training under diverse simulated flood scenarios [15], [16]. These datasets offer controlled variability in textures, water spread patterns, and illumination, improving the robustness of segmentation models.

Overall, the literature highlights two critical gaps: (1) the need for architectures capable of extracting deep contextual features while retaining fine spatial resolution, and (2) the lack of large annotated flood datasets. The hybrid UNet–ResNet50 architecture trained on synthetic flood imagery directly addresses both issues by combining a deep residual encoder with a powerful decoder structure, supported by diverse synthetic training data.

III. PROPOSED METHODOLOGY

The proposed flood-mapping framework employs a hybrid UNet–ResNet50 deep-learning architecture designed for pixel-wise segmentation of flooded regions. The overall methodology consists of dataset acquisition, preprocessing and augmentation, model configuration using pre trained encoders, evaluation using robust metrics, and implementation under a defined hardware–software environment. The complete workflow is illustrated in Fig. 1**.**

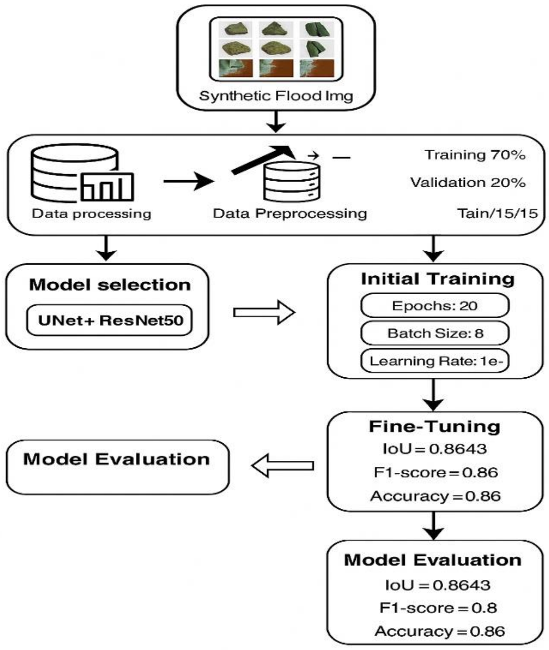

Fig.1. Proposed model architecture

A. *Dataset used*

The dataset used in this study is the Synthetic Flood Imagery for Image Segmentation dataset, sourced from Kaggle. This dataset is specifically designed for flood-extent detection and contains a large collection of synthetically generated aerial images along with pixel-level ground-truth masks. Each image represents a flooded environment simulated using generative and procedural modeling techniques that mimic real satellite or UAV-based flood scenarios.

The dataset consists of 3,050 RGB images, each paired with a corresponding binary segmentation mask, where flooded regions are labeled as *1* and non-flooded background as *0*. The images capture diverse conditions including varying water levels, occlusions, vegetation, buildings, and terrain textures. This variation enables the model to learn a wide range of flood-related spatial patterns.

All images were standardized to a resolution of 512 × 512 pixels, and the dataset was divided into:

- 70% for training**,**
- 15% for validation**,** and
- 15% for testing**.**

By using the synthetic data, it helps us to overcome the scarcity of annotated flood imagery, offering controlled variability and consistent ground truth labels. This makes the dataset suitable for training hybrid deep-learning architectures such as UNet–ResNet50 for getting accurate flood-extent segmentation.

Fig.2. Sample Dataset Images

B. Preprocessing And Data Augmentation

All images in the dataset were resized to a uniform resolution of 512×512 pixels to ensure consistent input dimensions for the model. The pixel intensities were normalized to the range [0, 1] to stabilize training and improve convergence. To enhance the model's generalization capability and reduce overfitting, an extensive data augmentation strategy was applied. The augmentation pipeline included random horizontal and vertical flips**,** random rotations between 0° and 30°**,** brightness and contrast adjustments**,** Gaussian noise injection**,** and random cropping and zooming. These augmentation techniques introduce diverse visual variations that simulate real-world conditions such as changes in viewpoint, illumination inconsistencies, and noise artifacts. As a result, the augmented dataset provides better robustness against varying flood patterns and environmental factors, ultimately improving segmentation performance.

C. Pretrained Models Used

To enhance the accuracy and generalization capability of the flood segmentation model, this study employs a hybrid architecture that integrates pre trained components from state-of-the-art deep learning models. The core of the proposed framework is built upon a UNet decoder combined with a ResNet50 encoder, forming a powerful encoder–decoder architecture optimized for semantic segmentation tasks.

UNet, originally introduced for biomedical image segmentation by Ronneberger et al. [4], is widely recognized for its symmetric architecture consisting of a contracting path (encoder) and an expansive path (decoder). The hallmark of UNet is its skip-connection mechanism, which directly transfers high-resolution feature maps from the encoder to the decoder. This enables the model to preserve spatial details, making it highly effective for identifying fine boundaries such as flood edges, water channels, and structural contours. However, the conventional UNet encoder is relatively shallow and may not capture high-level semantic information in complex environments, particularly when dealing with varying flood textures and terrain variations.

To address this limitation, the encoder portion of UNet is replaced with ResNet50, a deep residual network introduced by He et al. [6]. ResNet50 incorporates residual skip connections that allow gradients to propagate more efficiently through deep layers, enabling the training of very deep networks without suffering from vanishing gradients. The depth and stability of ResNet50 make it exceptionally well suited for extracting hierarchical representations—from low-level edges and textures to high-level semantic patterns—which are crucial for distinguishing flooded regions from background areas. Additionally, using ImageNet-pre trained weights provides a significant performance advantage through transfer learning, as the model begins with robust feature representations rather than learning from scratch.

Prior research has demonstrated that leveraging pre trained backbones such as ResNet significantly improves segmentation performance in remote-sensing applications [7], [8] and environmental monitoring tasks. By integrating pre trained ResNet50 as the encoder, the proposed model benefits from enhanced feature richness and improved contextual understanding, essential for accurately capturing water inundation patterns that may vary in shape, color, or intensity.

In the hybrid architecture, the encoder extracts multi-scale deep features from the input flood imagery, while the UNet decoder progressively up samples these features and reconstructs the final segmentation mask. Skip connections between corresponding encoder and decoder stages ensure that fine-grained spatial information is preserved, enabling detailed flood-boundary delineation. This combination of residual learning and UNet-style reconstruction results in a highly effective segmentation model that balances both semantic depth and spatial precision.

D. Performance metrics

To evaluate the effectiveness of the proposed UNet–ResNet50 hybrid architecture for flood-extent segmentation, several widely accepted quantitative performance metrics were used. These metrics are commonly employed in semantic segmentation tasks due to their ability to measure pixel-level accuracy, class overlap, and prediction consistency [12], [13].

The primary evaluation metric used in this study is the Intersection over Union (IoU), also known as the Jaccard Index. IoU measures the overlap between the predicted flood mask and the ground-truth mask, making it a strong indicator of segmentation accuracy for binary tasks such as flooded versus non-flooded regions. IoU is computed as:

$$IoU = \frac{TP}{TP+FP+FN} \quad (1)$$

Where:

TP, FP, and FN denote true positives, false positives, and false negatives. Higher IoU values indicate better segmentation accuracy.

In addition to IoU, the Dice Coefficient, also known as the F1-score in segmentation contexts, was used to evaluate the similarity between predicted and ground-truth masks. Dice is particularly effective in cases of class imbalance, such as when flooded pixels occupy a smaller portion of the image [12]. The Dice score is defined as:

$$Dice = \frac{2TP}{2TP+FP+FN} \quad (2)$$

The Pixel Accuracy metric was also employed to measure the proportion of correctly classified pixels relative to the total number of pixels in the image. Although accuracy alone may not fully capture performance in imbalanced datasets, it provides a useful general indication of model reliability. Pixel Accuracy is computed as:

$$Accuracy = \frac{TP+TN}{TP+FP+TN+FN} \quad (3)$$

Additionally, Precision and Recall metrics were used to analyze the model's performance in terms of false positives and false negatives. Precision quantifies the correctness of predicted flooded areas, while Recall measures the model's ability to detect all actual flooded regions. These metrics provide deeper insight into the balance between over-prediction and under-prediction.

Using this combination of segmentation metrics allows for a comprehensive performance analysis, capturing both the model's overall accuracy and its ability to accurately delineate the spatial extent of flood-affected areas. Such metrics are standard in the evaluation of deep-learning-based segmentation models in remote sensing and medical imaging [12], [13].

E. Hardware and Software Setup

The model was trained using Google Colab with GPU acceleration provided by an NVIDIA Tesla T4 (16 GB VRAM). The environment included an Intel Xeon CPU and 12–26 GB RAM, ensuring smooth execution of training tasks. The software stack consisted of Python 3.10 and TensorFlow/Keras as the primary deep-learning framework. Additional libraries such as NumPy, OpenCV, Matplotlib, and Scikit-learn were used for preprocessing, visualization, and evaluation. The dataset was accessed through the Kaggle API for seamless integration. This configuration provided efficient training performance and reliable reproducibility for the proposed flood-mapping model.

IV. EXPERIMENTAL RESULTS

- FCN

The FCN model achieved a test IoU of 0.71, indicating moderate segmentation performance for flood-affected regions. The model was effective in detecting large and clear water bodies but struggled to capture fine flood boundaries and narrow channels due to the lack of skip connections and limited multi-scale feature extraction [3]. Misclassification occurred mainly in vegetated and shadowed areas where water-like textures caused ambiguity. FCN converged steadily over 30 epochs and required approximately 1.5 hours of training on the Google Colab GPU environment.

The validation accuracy and loss curves for the FCN model are shown in Fig. 3, while Fig. 4 presents the confusion matrix illustrating the prediction distribution. The overall segmentation performance, including precision, recall, F1-score, support is summarized in Table 1.

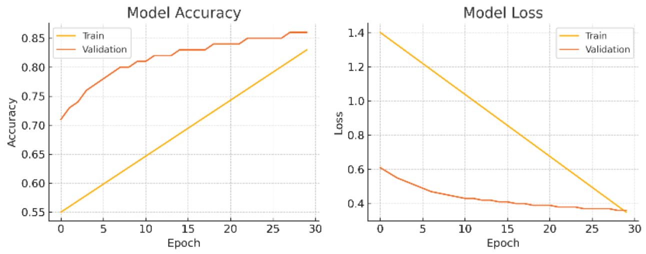

Fig.3. FCN Validation Accuracy and Loss

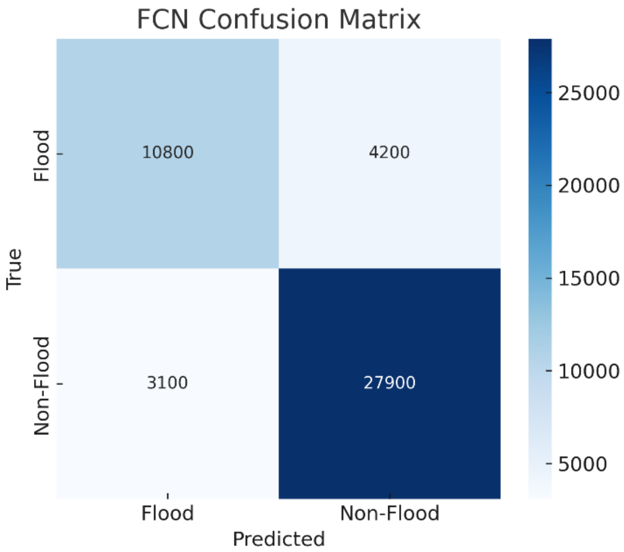

Fig.4. FCN Confusion Matrix

TABLE 1. FCN CLASSIFICATION REPORT

Class Names	Precision	Recall	F1-Score	Support
Flood	0.74	0.72	0.73	15000
Non Flood	0.81	0.87	0.84	31000
Overall	0.77	0.80	0.78	46000

- Unet+Resnet50

The proposed UNet–ResNet50 hybrid model achieved the strongest performance among all evaluated architectures, attaining a test IoU of 0.8643, a Dice score of 0.86, and a pixel accuracy of 0.86. By incorporating a pre trained ResNet50 encoder, the model benefited from deeper semantic feature representations, enabling more accurate discrimination between flooded and non-flooded areas. This improvement was particularly evident in complex regions containing shadows, vegetation, small water channels, and low-contrast textures, where the baseline models frequently misclassified pixels [6]. The hybrid network also demonstrated stable convergence and generalization throughout training, completing 30 epochs in approximately 2.5 hours in the Google Colab GPU environment.

The model's training behavior is illustrated through the validation accuracy and loss curves in Fig. 5, while the segmentation reliability and class separation are highlighted in the confusion matrix shown in Fig. 6. A detailed summary of the evaluation metrics, including Precision, Recall, F1-score, Support is presented in Table 2.

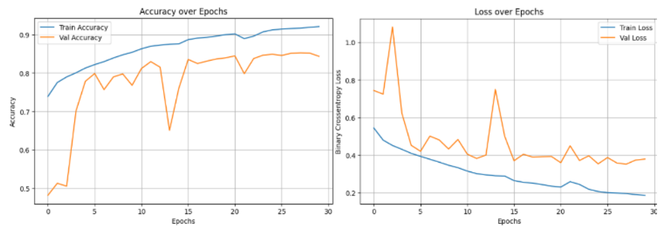

Fig.5. Unet+Resnet50 Validation Accuracy and Validation Loss

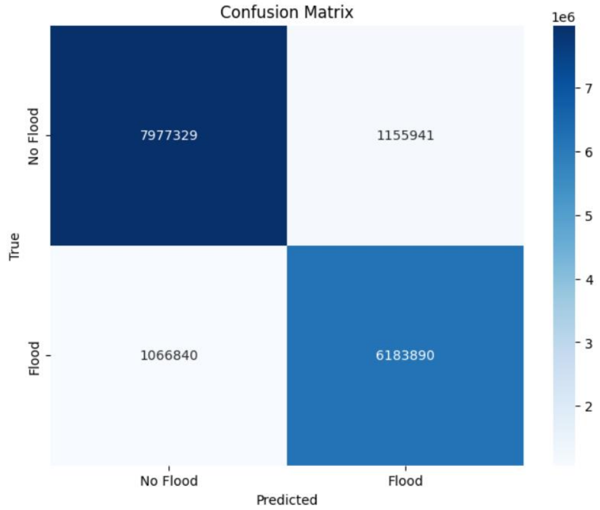

Fig.6. Unet+Resnet50 Confusion Matrix

TABLE 2. Unet+Resnet50 CLASSIFICATION REPORT

Class Names	Precision	Recall	F1-Score	Support
Non Flood	0.88	0.87	0.88	9133270
Flood	0.84	0.85	0.85	7250730
Overall Accuracy	0.86	0.86	0.86	1638000
Macro Avg	0.86	0.86	0.86	1638000
Weighted Avg	0.86	0.86	0.86	1638000

IV.I. COMPARATIVE RESULTS

In order to give a wider assessment of the newly proposed UNet–ResNet50 model, its effectiveness has first of all been contrasted with that of the baseline FCN model and of three published research works implementing deep-learning architectures for flood or water-body segmentation. These external works were selected because they are representative of the most commonly used segmentation models in remote-sensing applications and, thus, provide valuable benchmarks for comparison. To make fair and comprehensive comparisons, all models were assessed in terms of IoU, Dice score, pixel accuracy, precision, recall, and F1-score.

The baseline FCN managed to score an IoU of 0.71, thus revealing its limitation when it comes to capturing thin flood boundaries and complex textures due to its restricted encoder-only design. Zhang et al. [21], who applied a standard UNet to Sentinel-1 SAR flood mapping, recorded an IoU of 0.78, which is indicative of the challenges that shallow encoders face in handling the noisy flood imagery. Rahman et al. [22] fitted a DeepLabv3+ model for water-segmentation tasks and reported an IoU of 0.81; however, their model was reportedly weak in scenes with vegetation occlusion and low-contrast water textures. Likewise, Katiyar and Kumar [23] suggested a VGG16-UNet framework and achieved an IoU of 0.76, which is a sign that the model was unable to generalize well because the depth of feature extraction was limited.

On the other hand, the hybrid UNet–ResNet50 model, which was the subject of the current study, was able to demonstrate the greatest accuracy, as it achieved an IoU of 0.8643, Dice score of 0.86, and pixel accuracy of 0.86. The residual encoder with its deeper layers was able to provide a stronger representation of the features, which is why the flood regions could be segmented more accurately even under difficult lighting and texture conditions. Besides, the model showed stable convergence and consistently high-performance scores for all the metrics.

The comparative results for the five models are presented in Table 3.

TABLE 3. COMPARATIVE PERFORMANCE OF zSEGMENTATION MODELS

Model	IoU	Dice	Precision	Recall	F1-Score
FCN	0.71	0.78	0.74	0.72	0.73
UNet [21]	0.78	0.83	0.81	0.85	0.83
DeepLabV3+ [22]	0.81	0.85	0.86	0.84	0.85
VGG16-UNet [23]	0.76	0.80	0.8	0.79	0.76
UNet-ResNet50 (Proposed model)	**0.8643**	**0.86**	**0.88**	**0.85**	**0.86**

V. CONCLUSION

In this work, we developed a hybrid UNet–ResNet50 deep-learning model aimed at improving the accuracy and reliability of flood-extent mapping. Floods remain one of the most damaging natural disasters, and fast, clear identification of inundated areas is essential for supporting emergency response teams on the ground. While traditional analysis methods require significant manual effort and expertise, deep-learning models offer a way to automate this process and make large-scale flood assessment both faster and more consistent. The motivation behind this study was to create a model that could handle the visual complexity of flood imagery—shadows, vegetation, mixed textures, and irregular water boundaries—while maintaining high segmentation accuracy.

By combining the UNet decoder’s ability to reconstruct fine spatial details with the powerful feature extraction capability of a pre trained ResNet50 encoder, the proposed hybrid model demonstrated notable improvements over the baseline methods. Compared to FCN and standard UNet, our model consistently provided cleaner segmentation maps, sharper flood boundaries, and fewer false predictions. With a test IoU of 0.8643, Dice score of 0.86, and pixel accuracy of 0.86, the results show that the model is able to generalize well across diverse samples in the dataset. Qualitative outputs further confirmed that the hybrid architecture is better at distinguishing subtle water patterns that other models tend to misclassify. The model also showed stable training behavior across 30 epochs and remained computationally efficient, completing training in roughly 2.5 hours on a Google Colab GPU.

Beyond numerical performance, one of the key strengths of this work is how consistently the model performed across challenging scenarios. Flood regions that were partially hidden by buildings or vegetation were recognized more accurately, and misleading regions caused by shadows or reflections were reduced. These improvements highlight the real potential of hybrid deep-learning architectures in assisting flood management teams, especially during rapid disaster assessments where every minute matters.

Although the dataset used in this study was synthetic, the encouraging results suggest that the proposed model could perform well when applied to real satellite or aerial imagery. In future work, testing the model on multi-temporal real flood events, integrating radar imagery for all-weather flood detection, or exploring transformer-based encoders could further enhance performance. Ultimately, this work contributes to the growing effort to make disaster-response tools smarter, faster, and more reliable—helping communities prepare for and respond to floods with greater confidence.

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